SpartHill Hacks: Aviation Tracks

Sponsored by NCBAA
The Intelligent FBO & Every Gallon Counts

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Welcome aboard.

SpartHill Hacks brings builders together to turn ambitious ideas into working prototypes. NCBAA sponsors the aviation tracks, grounding the event in real business, general and commercial aviation challenges.

PURPOSE

Give participants and organizers a clear flight plan: two challenges, concrete deliverables and responsible-AI guardrails.



Two tracks. One operating principle:
AI assists. People decide.

Business / General Aviation
Prepare an FBO for Super Bowl LXI week in Los Angeles.

01 THE INTELLIGENT FBO

Commercial Aviation
Make supply loading leaner without compromising passenger service.

02 EVERY GALLON COUNTS

AI-POWERED TOOLS • PLANNING • SUSTAINABILITY • OPERATIONAL EFFICIENCY

The Intelligent FBO

Build an AI planning tool for the seven-day window around Super Bowl week in Los Angeles. Choose one operational pressure point and model it with public or synthetic data—documenting every assumption.



CHOOSE ONE FOCUS

- **AIRCRAFT ARRIVALS**
- **PARKING**
- **FUEL**
- **STAFFING**

From forecast to field plan.

BASELINE + FORECAST

Compare an ordinary week with Super Bowl week demand.

SCENARIO PLAN

Recommend preparation for low, medium and high demand.

MEASURE IMPACT

Show forecast accuracy and resource improvements.

JUDGING LENS

Innovation + accuracy / Practical preparation / Privacy, uncertainty and humans in command

Human in command.

Planning support only
No movement authority

Protect passenger privacy
No sensitive data

FINAL AUTHORITY

Explain uncertainty
Show assumptions

Stretch: adapt for
LA 2028 Olympics

What decisions should AI never make?



Every Gallon Counts

Build an AI tool for Cloudhopper Airways that optimizes either potable water or catering supply loading—reducing excess weight while protecting service and operational buffers.

SCOPE THE MODEL

1 supply category
1 aircraft type
3 routes

Use the provided synthetic dataset and state assumptions clearly.

Show the workflow. Prove the value.

WORKING PROTOTYPE

Planning workflow

3-MINUTE PITCH

User • problem • value

IMPACT + DISRUPTION

Baseline comparison and a demand spike or delay

ACCURACY + SAVINGS

SERVICE MAINTAINED

EXPLAINABLE + SAFE

Recommendation ≠ decision.

Every loading plan passes through a human approval gate.



What must remain human-only?

Organizer preflight.

01 Finalize briefs + synthetic datasets

02 Confirm judges + publish criteria

03 Prepare prototype + pitch submission

04 Clarify responsible-AI guidance

05 Schedule briefings + open Q&A

06 Coordinate track-specific support

Ready for takeoff.

THE INTELLIGENT FBO

Anticipate event-week demand. Turn forecasts into a practical, human-led preparation plan.

EVERY GALLON COUNTS

Reduce unnecessary weight. Preserve service through explainable, staff-approved loading plans.

NEXT: Confirm deadlines • Attend briefing • Build • Test disruption • Pitch

Thank you.

Questions and discussion welcome

Aviation track support team
Details and updates: SpartHill Hacks event portal

SPARTHILL HACKS × NCBAA

